c) Implementing Artificial Neural Networks (ANN)

Problem Statement:
Build a simple neural network to classify handwritten digits (MNIST).

Explanation:
ANNs are inspired by the brain and are used for classification, prediction, and learning.

Procedure:

1. Load data.
2. Define a neural network.
3. Train and evaluate.

Python Code (using Keras):

```
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
from tensorflow.keras.utils import to_categorical

# Load data
(x_train, y_train), (x_test, y_test) = mnist.load_data()
x_train, x_test = x_train/255.0, x_test/255.0

# One-hot encode labels
y_train = to_categorical(y_train)
y_test = to_categorical(y_test)

# Define ANN
model = Sequential([
   Flatten(input_shape=(28, 28)),
   Dense(128, activation='relu'),
   Dense(10, activation='softmax')
])

# Compile & Train
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
model.fit(x_train, y_train, epochs=5, validation_split=0.1)

# Evaluate
loss, acc = model.evaluate(x_test, y_test)
print("Test Accuracy:", acc)
```

Output:

```
Downloading data from
https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 1s 0us/step
/usr/local/lib/python3.11/dist-packages/keras/src/layers/reshaping/flatten.py
:37: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a
layer. When using Sequential models, prefer using an `Input(shape)` object as
the first layer in the model instead.
  super().__init__(**kwargs)
Epoch 1/5
```

```
1688/1688 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10s 5ms/step - accuracy: 0.8642
- loss: 0.4699 - val_accuracy: 0.9648 - val_loss: 0.1215
Epoch 2/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10s 6ms/step - accuracy: 0.9611
- loss: 0.1288 - val_accuracy: 0.9690 - val_loss: 0.1027
Epoch 3/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10s 6ms/step - accuracy: 0.9755
- loss: 0.0857 - val_accuracy: 0.9747 - val_loss: 0.0867
Epoch 4/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 9s 5ms/step - accuracy: 0.9822 -
loss: 0.0614 - val_accuracy: 0.9755 - val_loss: 0.0840
Epoch 5/5
1688/1688 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10s 5ms/step - accuracy: 0.9871
- loss: 0.0443 - val_accuracy: 0.9765 - val_loss: 0.0797
313/313 ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9730 -
loss: 0.0902
Test Accuracy: 0.9757000207901001
```